Part 2: Implementation Tasks

Q5

I use the video KandinskyBook.mp4, the top-left panel was replaced with a image I like (which is my favorite toy bear).

Steps

1. Locate the 4 Corners of the Panel
   I used OpenCV's cv2.setMouseCallback function to manually select the four corners of the top-left panel in the first frame of the video. This approach allowed precise annotation of the ROI without requiring complex corner-detection algorithms.
   Coordinates selected for top left: [[204, 537], [377, 536], [380, 761], [203, 763]]
2. Track the Corners
   I used the Lucas-Kanade Optical Flow algorithm (cv2.calcOpticalFlowPyrLK) to track the manually selected corners across subsequent frames. This method efficiently tracks feature points by estimating their movement between consecutive frames.
3. Compute Homography
   To calculate homography, I used OpenCV's cv2.findHomography, to compute a homography matrix for each frame. This matrix mapped the corners of the replacement image to the tracked corners of the panel in the current frame.
   The replacement image was warped to fit the panel using the homography matrix and OpenCV's cv2.warpPerspective.
4. Overlay
   The replacement image was seamlessly applied using a binary mask and bitwise operations:
   A mask was created for the warped image using cv2.fillConvexPoly.

The original panel region was removed from the frame with cv2.bitwise_and and cv2.bitwise_not.
The warped replacement image was added directly to the frame using cv2.add.
Without the remove of the original panel region, the brightness of panel region in the output version is too high. So, I use this way to ensure the replacement image retain its original brightness and color.

5. Output
   The modified frames were saved to an MP4 file (output.mp4) using OpenCV's cv2.VideoWriter

Challenges and Observations

The output video generally looks good to me but it has some noisy edges in some of the frames still.

1. Human error in manual corner selection
   Slight inaccuracies in manually selecting corners in the first frame could result in misaligned initial tracking, introducing noise and distortions in the warped image.
   Accurate selection of corners is critical to ensuring consistent tracking.
2. Edge Artifacts:
   The boundaries of the replacement image occasionally exhibited artifacts. This was likely due to:
   Subpixel inaccuracies in the homography matrix.
   Imperfect masking during the overlay step.
   Variability in the tracking algorithm's output for noisy frames.
3. Frame-by-frame processing.
   Frame-by-frame processing without feedback or reinitialization amplified small errors over time.

Interesting observations (/questions)

I found that when I directly overlay the replacement image onto the top left panel selection, I received a super bright version of replacement image in the output video. This is fixed by removing the original top left panel, I'm still not sure what caused the super bright change.